Urban Governance 5 (2025) 411–425

Contents lists available at ScienceDirect

Urban Governance

journal homepage: www.elsevier.com/locate/ugj

AI-powered meta analysis of geospatial participatory platforms for ecosocial transition in urban design and planning

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ARTICLE INFO

Keywords:
Collective Intelligence
Participatory design
Planning
ICT
Geospatial
Crowdsourcing

ABSTRACT

The ecosocial transition advocates for an approach that is ecocentric, transdisciplinary, resource-efficient, socially inclusive, equitable, and democratic. Over the past decade, digital participatory platforms (DPPs) have gained prominence in supporting ecosocial transitions by addressing chronic and wicked problems affecting both specific territories and the planet as a whole. However, fragmented overviews and biases in socio-spatial and socio-technical contexts have left a notable knowledge gap between technological advancements and research on their affordances in urban planning and design ecosystems.

This study examines how DPPs are introduced and conceptualized in literature. It explores distinguishing characteristics, emerging trends, and key success factors of geospatial DPPs through the lens of the collective intelligence (CI) framework. Following the PRISMA protocol, a systematic review was conducted across Scopus, Google Scholar, and public repositories, identifying 312 platforms worldwide. The coded descriptions of platforms were analyzed using content analysis, network mapping, and AI-powered clustering to reveal underlying patterns and relationships.

Our findings reveal a diverse spectrum of technological features, human-computer interactions, and engagement levels across DPPs. However, geographical distribution and effective institutional adoption remain limited. Nearly half of the examined geospatial platforms rely on one-way information flow focused on data aggregation, while co-creation platforms increasingly adopt advanced technologies such as gamified urban simulators, extended realities, and networked sentient technologies to foster interaction and co-design capabilities.

The insights from this comprehensive review contribute to a better understanding of DPPs for ecosocial transition, emphasizing key design aspects such as accessibility, transparency, transferability, user-friendliness, follow-up mechanisms, and continuous feedback. Given the complexity of urban governance and the challenges commonly highlighted in the literature regarding participatory approaches, DPPs are predominantly deployed to complement—rather than replace—existing top-down and technocratic planning frameworks. Strengthening the concept of participants as co-researchers within action research remains essential for fostering long-term systemic impact and stakeholder empowerment.

1. Introduction

Participatory design and planning (PD&P) roots lie in the ideals of participatory democracy, where decision-making is decentralised, pluralistic, transparent, and inclusive. Hence, participatory processes are crucial for solving wickled public issues and restoring community trust, as opposed to top-down policies (Laurian and Shaw, 2009). In addition, the PD&P process and outcomes can be enhanced by leveraging techno-social networks of collective intelligence (CI). In the PD&P literature, DiSalvo (2022) claim that CI is mostly an untapped resource to advance the participatory process and outcomes, while Sanoff (2007) defines it as a favourable outcome. Both long-standing concepts can be traced back as far as Plato's Republic, which emphasized principles such as 'freedom of speech, assembly, voting, and equal representation' (Sanoff, 2006) and Aristotle's Politics, which advocated for the idea that *'many heads are better than one'* (Dortheimer, 2022).

Today, participatory strategies are applied to urban planning, architecture, geography, policy, administration, public service, art, education, industrial and information technology (Luck, 2018, Sanoff, 2022), connecting an 'extended peer community'(Funtowicz and Ravetz, 1993). In particular, there are several rhyzomatic trajectories of PD&P across disciplines such as designing information technology as part of workplace democracy movements in Scandinavian countries (Björgvinsson et al., 2012, Ehn, 2008, Halskov and Hansen, 2015,

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https://doi.org/10.1016/j.ugj.2025.09.006
Received 4 January 2025; Received in revised form 17 July 2025; Accepted 16 September 2025
Available online 19 September 2025
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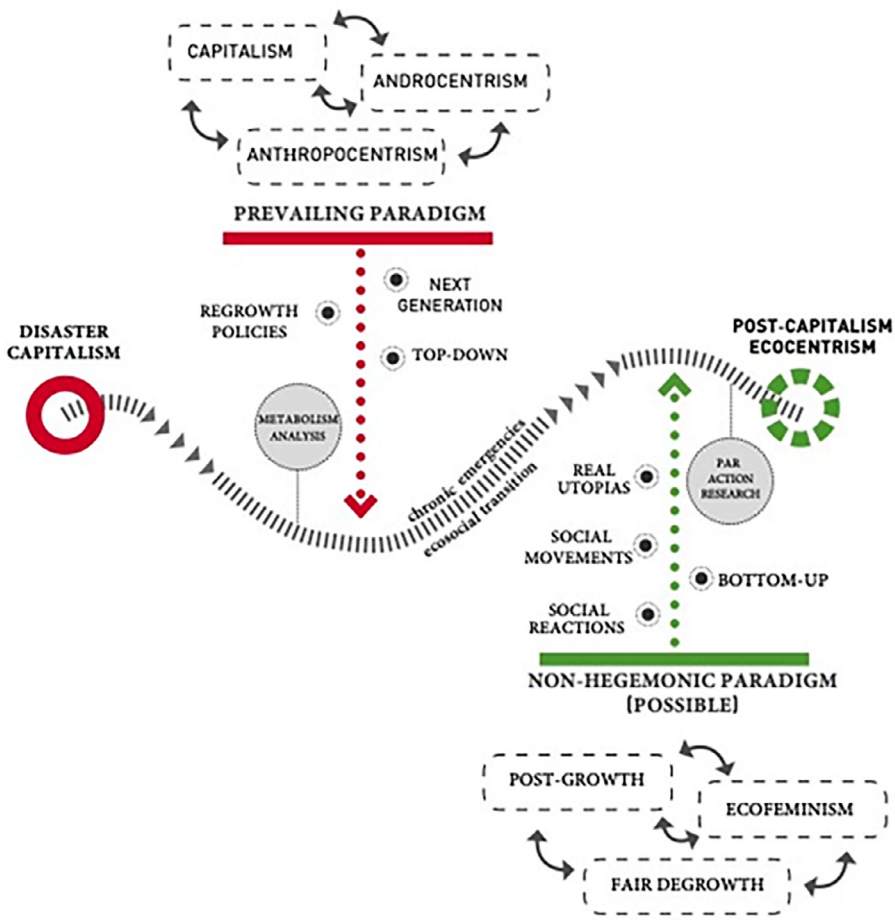

Fig. 1. Bottom-up and PAR within core topics for ecosocial transition (EMCROTUR).

Sanders and Stappers, 2008, Simonsen et al., 2013, Smith et al., 2017), advocacy design and planning services influenced by the 1960s civil rights movements and community consciousness in American society (Arnstein, 1969, Fagence, 1977, Godschalk, 1971, Sanoff, 1999), and renewed interests for architectural PD in the UK (Luck, 2018, Sanoff, 2007, Skeffington, 1969) almost half a century after grounding Design Participation Conference (Cross and Cross, 1972). Furthermore, government- and community-driven practices are emerging from institutions and civic movements addressing chronic issues, along with democratic innovations that promote new forms of participatory governance and politics in both the Global North and South (Newig and Rose, 2022, Pogrebinschi, 2021). Specifically, the Scottish charrette (Kennedy, 2017), Brazilian participatory budgeting (Sintomer et al., 2008), deliberative forums in China, Panchayati Raj reforms in India (Fung and Warren, 2011), North American and Western European universities' participatory design (Summers, 1979) and planning (Thornton, 1971) games, and eco-centric crowdsourcing applications are promising strategies for democratic innovation and ecosocial transition.

PD&P approaches are being applied across a wide range of practices in both developed and less developed countries under diverse allied concepts: cooperative, collaborative, diffuse, bottom-up, citizen-led, community-based, and grassroots (see glossary (Alarcón et al., 2022)). The combination of crowdsourcing and co-design strategies is referred to as distributed PD&P or mass-PD&P (Mueller et al., 2018). Positing that all these relevant design concepts are inherently collective and cooperative (DiSalvo, 2022), CI can advance the potential of diverse forms of participation and civic empowerment. Despite the growing literature on participatory collective paradigms and accompanying practices worldwide, there are few conceptual and theoretical frameworks on how to leverage CI through information and communication technologies (ICTs) for ecosocial transition in city-wide participatory processes. This study, part of an EU-funded project (EMCROTUR), conducts a meta-analysis of digital participatory platforms (DPPs) to support the project's bottom-up goals (Fig. 1) and widespread impacts through participatory action research (PAR).

PAR introduced '*research by the people*' based on the premise that user groups possess an expertise equal to, but different from, that of experts (Sanoff, 1999). By expanding the definition of who or what participates (Rice, 2017) based on the Latourian actor-network theory (ANT), it is also possible to include non-humans, such as objects, artefacts, and devices in the participatory process (Björgvinsson et al., 2012). Accordingly, ANT proposed ontological, methodological and epistemological turn for reconsidering PD&P with the techno-social aspects (Latour, 2005, Storni, 2015). Biocentric approaches, in particular, propose a more democratic paradigm shift by advocating for the inclusion of nature's rights in decision-making processes, especially concerning the commercialization, appropriation, exploitation, and privatization of vulnerable ecological urban areas. Serres (1995) emphasizes, how does nature speak, included as an actor (subject) in this new ethics? "*Indeed, the Earth speaks to us in terms of forces, bonds, and interactions, and that is sufficient to form a contract*". An ecological ethics with a new temporal dimension is presented (Vicente-Giménez, 2020, Riechmann, 2001), encompassing future generations and all species on the planet, to mediate between ecosystems and humanity and overcome the lack of governance in temporal matters. In sum, ANT drive us to reconsider both non-human abiotic actors in cyberspace (i.e., ubiquitous and interconnected infosphere) and non-human biotic actors within ecosocial transition in PD&P. This research addresses crowdsourcing geospatial platforms not just as innovations, but as actors actively reshaping the participatory settings.

Despite the availability of promising repositories and reviews, there is a pressing need for up-to-date research on the expanding use of DPPs in PD&P, specifically geospatial crowdsourcing platforms. The above-mentioned reviews within specific socio-spatial contexts and technological features provide a valuable foundation for further assessment of DPPs. In pursuit of this, we created a network visualization to comprehensively map DPPs across the reviews and repositories (Fig. 5), identifying overlaps and assessing their diversity in design empowerment. Since the exhaustive inventory introduced by Falco and Kleinhans (2018), based on a systematic review conducted in November 2016, an additional 1,700 documents have been added to Elsevier's Scopus through 2024. Falco and Kleinhans (2018) also confirmed that the research is lagging behind the increased deployment of DPPs. In grey literature, PD&P practices have also expanded significantly; for example, the Participedia database grew from 722 cases (completed before 2017) to 2,263 cases by October 2024. The number of DP platforms, along with platformization processes and their impact on social media, governance, and planning, has grown, particularly during the recent pandemic. This continuous expansion of (grey) literature highlights the importance of updating the review of DPPs from a novel CI perspective.

This paper aims to deepen the understanding of how DPPs can facilitate advanced PD&P settings in relation to ecosocial bottom-up transition, in contrast to technocratic top-down approaches primarily rooted in regulation, legitimization, and control. It also aims to analyze the methodological, contextual, and actor-related dimensions through the CI genome, identifying key characteristics, success factors, barriers, and enablers across various platform settings. To achieve this, we conducted a literature review encompassing peer-reviewed publications indexed in the Scopus database and grey literature, including three open-access repositories. Rather than evaluating the platforms themselves, this study focuses on how platforms are conceptualized, framed, and applied in literature, particularly through reported uses in empirical case studies. However, the numerous and ever-increasing platforms pose a challenge to analyze and necessitate a well-defined scope and framework. To investigate different ways of harnessing CI, we utilized foundational works and prevailing guidelines, with a particular focus on geospatial crowdsourcing DPPs. These approaches are closely linked to the promises of CI, citizen design science, voluntary participation, and soft GIS strategies in mobile and web-based environments. Framing our research within this context, we formulated the following research questions (RQs):

1. How are DPPs conceptualized in relation to ecosocial transitions in PD&P, and what patterns and trends emerge from the literature?
2. What enablers and barriers are reported across the DPP lifecycle, and how do key design aspects drive successful engagement?
3. What affordances and emerging novel forms are attributed to geospatial DPPs?

By addressing these RQs, our review provides a comprehensive understanding of the affordances of DPPs, offers insights for designing efficient PD&P settings, and sparks new RQs for future research.

2. Background

2.1. Leveraging collective intelligence for ecosocial transition

Contemporary urban planning and architecture develop under the influence of urban-tourism growth mechanisms, which lead to economic, social, and ecological issues amid a chronic global crisis. The growth they promote is driven by a tacit agreement encouraged by politicians, businesspeople, media, and professionals as the main interested parties (Logan and Molotch, 1987). The oversized urban metabolism exacerbates crises related to greenhouse gases, biodiversity, energy, inequality, and democracy, posing a barrier to a just ecosocial transition.

Modernity culture pursues progress and promotes growth as a means to achieve it (Meadows et al., 1972). Instead, the ecosocial transition challenges the disciplines of architecture and urban planning to radically rethink their ideas, methods, and practices. This is due to the clear incompatibility between physical and economic growth and the ecosocial goals of a just paradigm, both for specific territories and the planet as a whole.

Degrowth theory offers a framework that reorients the growth-driven system—viewed as anti-systemic—toward a systemic paradigm based on reducing economic metabolism, particularly in developed regions. In this context, degrowth theory and post-growth practices advocate for lowering constructive and territorial metabolism and promoting ecological adaptation through a process defined as a just ecosocial transition (Latouche, 2013). In this way, the ecosocial transition proposes a new approach that is more ecocentric, transdisciplinary, resource-efficient, socially inclusive, equitable, and democratic (Fig. 1).

In response to emerging chronic crises in urban governance and planning, bottom-up approaches promoted by social movements and participatory mechanisms seek to advance equitable and democratic ecosocial processes. However, academia has yet to analyze or characterize the few existing practices and opportunities that could serve as replicable models. As part of a broader investigation covering stakeholder perspectives and evidence from participatory cases (Heyik et al., 2024), this research examines DPPs to address these gaps. All of this presents an opportunity for a radical rethinking of the current system and the adoption of the ecosocial transition paradigm in design and planning, particularly in the most oversaturated and exploited territories, which are home to populations most affected by climate change, energy scarcity, social inequalities, and lack of democracy.

Ecological degradation, social inequalities, and health crises present long-term challenges for policymakers, citizens, researchers, and designers (Kimbell et al., 2022). Participatory strategies promote sustainability principles aligned with the UN Sustainable Development Goals (SDGs), fostering just, sustainable, and democratic communities. Costanza-Chock (2020) lists ten principles of Design Justice Network, and many of them related with the CI promises. By involving more people, data, and technology, CI can bridge five critical gaps in data, action, distance, diversity, and decision-making (Berditchevskaia et al., 2024).

CI offers benefits such as identifying root causes, improving awareness, enabling inclusive decisions, strengthening legitimacy, ensuring accountability, and supporting civic monitoring (Peach et al., 2019, Ryan et al., 2020). Institutions like MIT Center for CI, Nesta, GovLab, and the Co-Intelligence Institute have designed and deployed CI strategies to tackle global challenges. EU-supported projects, including CIPTEC, C3PO, Cities4People, U_CODE, and COMRADES, have developed CI tools aligned with SDGs. For example, Catalyst facilitates debates, reflection, and solutions to complex problems (Arniani et al., 2015). CI has supported movements such as NIMBY, feminist, big society, activist, maker, do-it-yourself urbanism, localism, ecocentric, and open governence. While tools like Co-Cities, OECD, and UN-Habitat's civic engagement index (SDG 11.3.2) measure engagement, a comprehensive framework for global cities remains absent.

2.2. Digital participatory platforms

Many organizations, like Participedia, LATINNO, Partscout, Co-Cities, UserCentriCities, UN-Habitat, Deliberative Wave, Co-Governance, TROPICO, CitizenScience, and CivicWise, have evaluated and shared PD&P cases, but few focus on DPPs. Unlike technocratic planning support systems (PSSs), DPPs enable collaboration and empower people in decision-making and co-creation. PeoplePowered, a non-profit organization, evaluate 30 multipurpose and specific DPPs, first launched in 2022. COLDIGIT, Collective Intelligence through Digital Tools involving five organisations from Scandinavia and the UK, presents 71 tools related to public participation and CI. Furthermore, previous reviews of such platforms have examined different aspects linked to epistemic objectives (e.g., harnessing advanced technologies like AI-powered strategies), and pragmatic objectives (e.g., comparing

Table 1
Overview of researches on DPPs with key variables and methodologies (created by Authors).

Authors	Model	Data Collection	Selection Criteria	DPPs & Cases	Distribution	Analysis
Shin et al., 2024	Collective Intelligence Genome	Public repositories (PeoplePowered, OECD, and COLDIGIT) Nov–Dec 2022	Popular or better-known items in 2022 recognised by field experts Current State	116 DPPs	Worldwide	Hierarchical clustering
Fornaroli & Gatica-Perez, 2023	PRISMA framework for urban crowdsourcing platforms	Systematic review in Scopus, ACM, Wiley, SRRN, and IEEE October 2021	Type and purpose Official status and link to the city Current state	32 DPPs	Worldwide (~70% of the platforms in Europe/North America)	Content analysis
Renteria & Varela-Castro, 2023	A conceptual framework of a crowdsourcing platform's architecture with four decision points	Systematic review for typology of platforms Cases obtained from a public repository (CrowdLaw)	Gov-led platform Online crowds' contributions Currently active website between Oct–Dec 2019	52 DPPs	Worldwide (~85% of the platforms in Europe/ America)	Hierarchical clustering
Babelon et al., 2021	SPP/IAP2 International Association for Public Participation Spectrum	Online survey with local government agencies (December 2017 - January 2019)	Web-based Geo-participation Gov-led planning Diverse scales Timescale (2015-18)	9 DPPs 25 Cases	9 developed countries in Europe, North America, and Australia	Thematic analysis focuses on empirical data
Türken & Eyüboğlu, 2021	Adapted framework comprises project, participatory, and socio-technical criteria	Narrative review	E-participation The upper levels of participation Location-based Citizen design	10 DPPs (5 location-based and 5 citizen design)	Not defined	SWOT
Kahila-Tani et al., 2019	Challenges of participatory planning process: effectiveness, representativeness, and knowledge quality	Online survey with practitioners and professional planners at the Metrex Conference 2017	Using Maptionnaire Min. of 20 participants Formal urban and regional planning Timescale (2014-17)	1 DPP 203 Cases	Cases in Finland (62%) and other countries (38%)	Critical analysis pros & cons
Gün et al., 2019	Six instances of design empowerment, funcionalities, objectives, and design action phases	Systematic review in Web of Science and Scopus databases Exploring research projects and G.Scholar	Empowerment levels Exclusion of one-sided information exchange (lowest level) European cities	25 DPPs	Europe	In-depth content analysis
Gil et al., 2019	Arnstein ladder of participation	DPPs selected in different cities and smart governments across different countries	Testing e-governance approaches Active (in early 2016) Innovation across cities and countries	13 DPPs	Worldwide	Typology-based
Falco & Kleinhans, 2018	Typology of citizen government relationship: technological features, kinds of problems, and pricing patterns	Systematic review in Scopus, Google search, and the list of blog 'communitymatters' November 2016	Not a government portals (informing) Co-production Currently active Neighbourhood matters (self-org)	113 DPPs 25 co-production DPPs and related cases	21 countries around the world	Critical overview and detailed typology-based analysis
Ertiö, 2015	Four quadrant model of Participatory Apps (type of data and information flow)	Web portals, social media, distribution lists, and via discussing with experts & practitioners 2013-2014	M-participation Environment- and people-centric Relevance for planning Multiple roles Transferability	35 Apps	Worldwide (from the US to Hong Kong, and from Australia to Finland)	Typology-based
Desouza & Bhagwatwar, 2014	Four archetypes of DPPs: dominant actors and the pattern of information flows	Secondary data (e.g., articles, press, and other media mentions)	Technology-enabled Exclusion of information conveyance DPPs	25 DPPs	16 most populated US cities	Exploratory Typology-based

DPPs' affordances and utilities) in urban planning and governence. Table 1 summarizes research on DPPs, highlighting key variables, models, data collection techniques, selection criteria, the number of platforms, their geographical distribution, and analysis types, conducted to identify knowledge gaps in the literature.

Existing reviews analyzed DPPs from different perspectives, including design empowerment levels (Gün et al., 2019), dominant actors and information flows (Desouza and Bhagwatwar, 2014), decision points (Renteria and Varela-Castro, 2023), and citizen-government relationships (Falco and Kleinhans, 2018). Ertiö (2015) identified approximately one hundred DP mobile apps through web portals, social media, distribution lists, and discussions with experts and practitioners in 2013-14, of which 35 were analyzed. In a seminal paper, Falco and Kleinhans (2018) analysed and classified 113 active DP platforms within a citizen-government relationship typology. Gün et al. (2019) identified 106 web-based and/or mobile DP platforms in Europe, select-

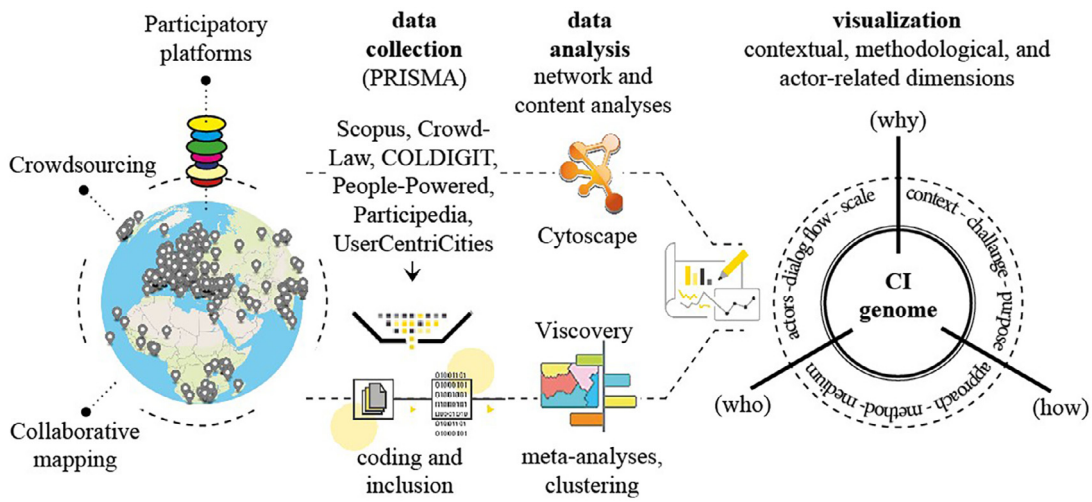

Fig. 2. Research design (created by Authors).

ing 25 by excluding those limited to one-sided information exchange. Gil et al. (2019) surveyed 13 DP platforms that connect citizens, government, and businesses in smart cities and governance. Türken and Eyüboğlu (2021) examined 10 platforms representing higher levels of participation. Babelon et al. (2021) presents empirical evidences based on the opinions of planning experts to assess 9 DP platforms deployed use-cases in cities across Europe, North America and Australia. Renteria and Varela-Castro (2023) analyzed patterns in the technical design of 52 DP platforms from CrowdLaw catalog. Fornaroli and Gatica-Perez (2023) conducted a systematic review and collected 32 urban crowdsourcing DP platforms. Shin et al. (2024) used the CI genome framework to examine 116 DPPs from COLDIGIT, OECD, and People-Powered databases in November 2022. Overall, these reviews collectively provide a useful, diverse, and reliable foundation for assessing and enhancing the ecosystem of DPPs. Building on a critical assessment of these earlier studies, we developed a methodological framework specifically designed to address the key conceptual and analytical gaps identified in the literature.

3. Method

Several meta-analyses have mapped the PD&P landscape, each focusing on different areas but sharing the goal of enhancing understanding and generating new inquiries (Beierle and Cayford, 2002, Newig et al., 2019, Tuhkala et al., 2018). Our research design consists of two main stages: first, identifying relevant case studies from the literature; and second, analyzing how these DPPs are described and conceptualized, using the CI framework.

To achieve this, we followed the PRISMA protocol to systematically identify platform cases in academic and grey literature. Importantly, this study does not assess the real-world performance of DPPs, but rather analyzes how they are represented and framed within academic and grey literature. Clustering techniques were applied to identify recurring patterns, while content analysis was used to examine the affordances and design characteristics attributed to these platforms. The proposed model follows the four basic procedures of meta-analysis:

1. Identify relevant DPPs in the literature according to selection criteria
2. Develop a coding framework for CI genome
3. Code the platform descriptions based on the attributes, using multiple raters.
4. Analyze the genome variables to uncover patterns within the DPPs, considering RQs.

To avoid fragmented overviews and selection bias, and to ensure a comprehensive inclusion of DPPs, we conducted a multi-source literature search in both English and Spanish. Scopus was selected as the primary academic database due to its wide interdisciplinary coverage, advanced filtering options, and compatibility with PRISMA protocols. To expand the scope and include grey literature and non-indexed sources, we also searched Google Scholar and three public repositories relevant to urban innovation and digital participation.

In particular, we took into account the systematic review conducted by Falco and Kleinhans (2018), which used Scopus, Google Search, and the blog list *communitymatters*, covering literature up to 2016. Building on their work, our review focuses on studies published from 2016 onwards in order to address more recent developments in digital participation.

The search keywords and selection criteria are detailed in the research design (Fig. 2) and structured according to the PRISMA framework. To strengthen the analytical dimension, we applied AI-powered clustering using Viscovery and performed network analysis using Cytoscape.

The aim is to both update and complement existing findings while advancing the field through the application of a new analytical technique—Self-Organizing Maps (SOM)—to identify emerging patterns and typologies within DPPs.

3.1. Identification of DPPs

The inclusion criteria in previous reviews are included in Table 1. We defined our selection criteria to reach a comprehensive (global) and diverse DPPs. Eligibility based on the following critera:

- Relevance for PD&P
- Versatile levels of engagement (specific or multi-purpose use cases)
- Geospatial application (GIS-powered)
- Ecosystem of participation (gov- and/or citizen-led)
- Global North and South (geographical distribution)

Our focus was on platforms that enable interaction and collaboration between actors and utilize novel CI strategies. We excluded government-to-business (G2B) and government-to-government (G2G) platforms, as well as one-way informing/transparency platforms. We also excluded platforms that, despite initially exploring citizen participation, later shifted their focus and did not progress from the prototype stage to real-world application (Fig. 3).

3.2. Evaluation framework

To evaluate DPPs, we developed a three-dimensional analytical framework, referred to as the Collective Intelligence (CI) genome. This

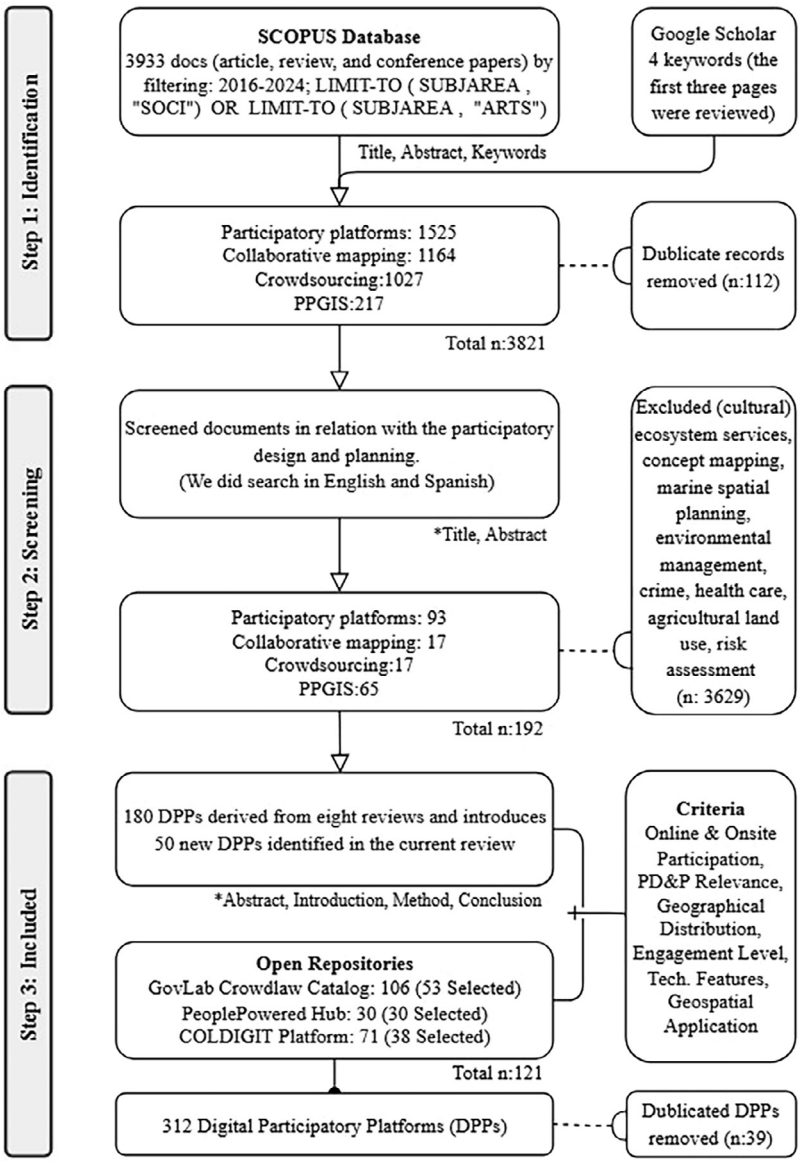

Fig. 3. PRISMA framework for the identification and selection of DPPs (review conducted in October 2024).

framework builds on prior participation models and enables a structured and comparative understanding of diverse DPP cases. It integrates contextual drivers (why), methodological characteristics (how), and actor-related configurations (who), inspired by existing evaluation studies, participation taxonomies, and CI conceptualizations.

Several studies have provided foundational assessment criteria. Lin and Benneker (2022) evaluated Planning Support Systems (PSS) in Finland using criteria such as user-friendliness, usefulness, and efficiency. Rzeszewski and Kotus (2019) used system usability scales (SUS), cognitive load (CL), and eye-tracking to assess geospatial platforms in participatory planning. Ballatore (2020) introduced the Participatory GIS Usability Scale (PGUS), assessing user interface, spatial interface, learnability, effectiveness, and communication. Similarly, Rico et al. (2022) consolidated various evaluation models and proposed three main criteria: user experience, learning, and effectiveness.

To understand public empowerment, we refer to Arnstein's and Rocha's ladder of participation and IAP2's spectrum (inform, consult, involve, collaborate, and empower). The Horizon 2020 project U_CODE (Urban Collective Design Environment) adapted this into a four-level model by excluding 'involve' and clarifying information flows at each stage:

- One-way (authorities to citizens) in inform,
- Limited two-way in consult,
- Advanced two-way in collaborate,
- One-way (citizens to authorities) in empower.

Ramos et al. (2019) also proposed five engagement categories—curated, contributory, interactive, facilitated, and constitutive—to define the degree of agency afforded to participants. Senbel and Church (2011) introduced the 6-I framework for design empowerment: information, inspiration, ideation, inclusion, integration, and independence.

In addition, CI applications operate using various methods and technologies, including computer vision, crowd forecasting, crowdmapping, microsurveys, natural language processing (NLP), peer-to-peer exchange, predictive analytics, and remote sensing (Peach et al., 2021). Kanhere (2013) categorizes mobile participatory sensing apps into people-centric (collecting tacit knowledge and spatial preferences) and environment-centric (co-sensing through sensor and observational data). Ramos et al. (2019)) further distinguish engagement modes as: play, immerse, sense, create, and deliberate. The Catalyst project (Arniani et al., 2015), built on web and networked ICTs, defines CI dynamics along two modes—aggregation and co-creation—reflecting levels of awareness and civic intelligence.

We integrate these models into CI genome (Malone, 2018), combining Nesta's people–data–technology lens, Kanhere's (2013) participatory sensing typology, and IAP2's participation spectrum to reveal insights through the indentified DPPs. The resulting framework supports systematic analysis of DPPs via a triple-scale coding scheme, aligned with prior studies and allowing comparative insights.

Dimension 1: Why?

This dimension investigates the underlying rationales for the development and deployment of digital participatory platforms (DPPs), focusing on the societal challenges they aim to address, the goals they pursue, and the thematic agendas they reflect.

- **Main Topics:** These themes were derived from a systematic review of literature and grey sources, based on recurring keywords and concepts. The most common topics include: Responsive City, Citizen Design Science, ICT in Smart Cities, Gamification in Urban Planning, Sustainable City, Collaborative Planning, Citizen-responsive planning, Feminist Urban Planning, Participatory Greening, Participatory AR in Public Space, Augmented Collective Intelligence.
- **Level of engagement:** Engagement levels are categorized as inform, consult, involve, collaborate, and empower, in alignment with IAP2's participation spectrum and the U_CODE model (Senbel and Church, 2011). These levels represent increasing degrees of public influence in DPP ecosystem.
- **SDG relevance:** The societal impacts of the platforms were also examined in relation to the United Nations Sustainable Development Goals (SDGs). The benefits align with the following targets: new forms of accountability and governence (SDG16), anticipating, monitoring and adapting to sytemic risks (SDGs 11, 14, 15), real-time monitoring of the environment (SDGs 11, 15), understanding and working with complex systems (SDGs 10, 11, 12), Inclusive development and technologies (SDGs 5, 10), distributed problem solving (SDGs 2, 3, 13) (Peach et al., 2021).

Dimension 2: How?

This dimension examines the operational logic of digital participatory platforms (DPPs), focusing on how they function in practice—through interaction models, technological configurations, modes of data exchange, and interface environments that structure user engagement.

- **Approach:** Aggregate or collaborate (based on Catalyst and U_CODE)
- **Information flow/exchange**: one-way or interactive, based on typology of participatory apps of Ertiö (Ertiö, 2015)
- **Type of data collected** (Kanhere, 2013): environment- and citizen-centric
- **Medium**: mobile, web-based, extended reality, and hybrid (projection mapping, AR, object tracking, and touchscreens)
- **Technological features** (Falco and Kleinhans, 2018, Gün et al., 2019): Map-based comment/idea tools, multimedia uploads, voting and ranking options, collaborative mapping, analytics, crowdfunding, surveys, geo-budgetting, drawing polygons on maps, designing 3D objects, and follow-up mechanism

Dimension 3: Who?

This dimension investigates the actors involved in the development and implementation of DPPs, as well as their user groups and spatial deployment—from local artifacts to global applications—highlighting sectoral responsibilities and geographical distribution across different world regions.

- **Developer/Provider** (Shin et al., 2024): first (public) sector, second (private) sector (e.g., tech firms), and third (voluntary) sector (e.g., NGOs and universities)
- **Spatial scale**: Artifact → Local → Citywide → Regional → Global
- **Geographical distribution** (Falco and Kleinhans, 2018): Representation across Global North and South—Asia, Africa, Oceania, Europe, the Americas

Through this CI genome, we aim to uncover common patterns and emerging trends among DPPs. The triple-scale framework allows a nuanced analysis of motivations, mechanisms, and actors, contributing to a robust and transferable model for evaluating participatory digital platforms in urban planning and beyond.

A key limitation observed in previous applications of dimensional frameworks with clustering algorithms is the tendency to assign equal weight to all variables, regardless of their relevance or discriminatory capacity. This uniform treatment may obscure the influence of critical parameters or overemphasize less informative ones. In contrast, our approach incorporates Self-Organizing Maps (SOMs), which allow for the grouping and prioritization of variables through built-in functionalities such as priority weighting and variable grouping. These features enhance the robustness and interpretability of clustering outcomes.

Accordingly, the evaluation framework presented in this study is not intended as a fixed or prescriptive model, but as a flexible analytical foundation that can be extended in future research by incorporating new dimensions or adjusting the relative weights of variables to suit different contexts and research objectives.

4. Results

The identified platforms were analyzed based on engagement levels within network analysis, incorporating insights from previous review studies. This descriptive analysis presents findings from both study-centric and platform-centric perspectives. Clustering analysis was conducted using Self-Organizing Maps (SOMs, also known as Kohonen maps) in the Viscovery software to uncover patterns across identified DPPs.

4.1. Descriptive results

The included studies show the highest contributions from the United States, Finland, and the United Kingdom, followed by Italy, Spain, and Australia. Other notable contributors, such as Germany, Poland, and Austria, reflect a strong Global North representation, resulting in fragmented spatial coverage and limited applicability beyond these contexts. The Global South accounts for approximately 10% of the studies reviewed in the second step. However, by combining existing reviews and public repositories, we covered a more diverse range of countries than any previous study. Document distribution shows steady growth until 2019, a peak in 2020 due to the pandemic, and stabilization at around 20–22 documents in 2023 and 2024.

Following step 2 of the PRISMA protocol, 192 studies were screened based on their abstract, introduction, method, and conclusion parts. Platforms were excluded due to duplication with existing literature, lack of data beyond the prototype stage, or inability to establish PD&P relevance. Of the 50 new DPPs identified, 60% employed qualitative analysis, 15% conducted quantitative analysis, and 70% centered the platforms in the core of the paper.

4.2. Visualizing networks

In the platform-centric findings, a network analysis was conducted on the 262 DPPs (50 platforms from our review were excluded for graph clarity) examined in existing reviews (Fig. 4). These reviews utilize various models, including CI, IAP2, the typology of citizen-government relationships, the Four Quadrant model of participatory apps, and socio-technical aspects. Data collection methods included research projects, discussions with practitioners and planners, online surveys with local government agencies, utilization of public repositories, and narrative/systematic reviews. The geographical distribution of DPPs is dominated by the Global North, specifically Europe and America. Methods such as critical overview, exploratory typology-based analysis, in-depth content analysis, SWOT, thematic analysis, and clustering were employed. A comparison of key variables reveals significant patterns,

Fig. 4. Level of empowerment for 262 DPPs from reviews and public repositories (created by Authors).

such as larger sample sizes in clustering analyses (50–120 DPPs) and highlights the need for further work to consolidate existing reviews and expand the current list of DPPs.

Fig. 4 and Table 1 are two key sources to be read together. When examining the key variables within this network, important insights can be drawn. Platforms covered in multiple reviews, i.e., intersecting DPPs, are highlighted in red. Additionally, in this analytical networking diagram, the level of engagement for DPPs is represented by geometric shapes: triangle for inform, diamond for consult, square for collaborate, and circle for empowerment. All attributes within the CI genome framework can be visualized within the network. These networks allow us to understand the DPP ecosystem across different socio-spatial and socio-technical contexts and identify different approaches. Falco and Kleinhans (2018) classified DPPs into a single category using the citizen-government relationships model, whereas Babelon et al. (2021) allowed respondents to assign multiple categories based on the IAP2 Spectrum of Public Participation model. Similarly, Gün et al. (2019) classified DPPs according to their highest level within the six instances of design empowerment model.

- MinStad and Maptionnaire are examined in both reviews (Falco and Kleinhans, 2018, Gün et al., 2019). The first categorizes them under the interaction level, while the second evaluates them as design-empowering aspects at the independence level.
- Fix My Street is examined in both reviews (Falco and Kleinhans, 2018, Gil et al., 2019). The first categorizes it under the interaction level, while the second evaluates it at the informing level within the scope of B2G, G2C, and C2G (B: Business, G: Government,

C: Citizen). Similarly, Ushahidi is categorized under the interaction level in the first review, while it is evaluated at the citizen control (empowerment) level in the second review.

- CityLab010 is evaluated at the inclusion level (Gün et al., 2019), whereas it is also classified within the co-production level (Falco and Kleinhans, 2018).
- Carticipe is listed as a platform for co-producing ideas and proposals (Falco and Kleinhans, 2018), assessed in the consult category (Babelon et al., 2021).
- While Emotional Maps categorized under informing level within the study of Falco and Kleinhans (2018), the results of the survey (n:28) conducted by Babelon et al. (2021) indicate the versatile engagement levels (inform-3; consult-2; involve-4; collaborate-3; empower:3) for the same tool.
- Babelon et al. (2021) examined nine DPPs across 25 cases, illustrating that geoparticipation objectives in urban planning have versatile uses. These tools can facilitate multiple levels of engagement simultaneously; for instance, Commonplace, Emotional Maps, and Social Pinpoint platforms support objectives ranging from consultation to collaboration.

Versatile engagement can vary based on the diverse 'evaluation frameworks' discussed in Section 3.2. As a result, distinct approaches are observed in applications—systems tailored for specific areas, such as citywide ePart.it, Tirana, and Next Bangalore; project-based platforms like Qua-kit and Transformcity; and adaptable mapping platforms like Ushahidi, designed for diverse issues and communities. Some platforms serve a single objective with a rigid structure, while others fulfill multiple objectives with an intuitive and flexible components.

4.3. Clustering features

An AI-powered clustering technique was employed to identify common patterns based on the coded CI attributes, using the evaluation framework detailed in Section 3.2. SOMs provide a simpler method for clustering multi-dimensional mixed data with minimal preprocessing (Ritter and Kohonen, 1989). The SOM was trained with 1000 nodes, automatic map format, Ward clustering, equal attribute priority (1), tension 0.5, temperature 0.2, and a normal schedule. Approximately half of the platforms were included in the clustering analysis, deemed sufficient for the initial phase. Future studies will focus on expanding the sample and refining genome variables for further analysis. The SOM-Ward clustering reveals distinct characteristics across six clusters, with varying attribute concentrations and interconnections highlighting distinguishing features (Fig. 5). Variables such as geographical distribution, pricing, provider sector, and years of deployment were excluded from the clustering results due to their diffuse distributions and limited explanatory power. Instead, variables with stronger clustering patterns and relevance to platform affordances were selected to enhance understanding through cross-analysis.

Each cluster reflects the influence of multiple variables and open for new attibutes for further analysis. In particular, examining the key design aspects related to RQ3 for each cluster provides deeper insights:

- The first cluster (C1) focuses on **deliberative** platforms, enabling interactive contributory and constitutive processes through citizen-centric data (opinions, votes, ideas, etc.) aggregation and decision-making features in a hybrid medium (web and mobile) with 2D mode (such as geo-budgeting platforms).
- The second cluster (C2) focuses on **analytical** platforms, enabling both one-way curated and interactive contributory processes through citizen- and environment-centric data aggregation in diverse mediums with 2D mode (such as citizen science monitoring and participatory sensing platforms).
- The third and largest cluster (C3) focuses on **crowdsourcing** platforms, enabling interactive and collaborative processes through citizen- and environment-centric data aggregation in diverse mediums with 2D mode (such as collaborative mapping platforms).
- The fourth cluster (C4) focuses on web-based **co-creative** platforms, enabling collaboration between agents in 2D mode (such as gamifying and micro-tasking platforms like QurbanCraft and Architasker).
- The fifth and smallest cluster (C5) focuses on **immersive** platforms, enabling collaboration between agents in an XR medium with 3D mode (such as Urban Gallery-DLC, 4D Citymaking, and Veus).
- The last cluster (C6) focuses on **co-design** platforms, enabling two-way interaction between agents and covering diverse technological features in 2D and 3D modes (such as EquiCity and CoDAS).

Expanding variables, sample size, and adopting improved techniques can enhance the reliability of findings. In the discussion, results are compared with Shin et al. (2024) and Renteria and Varela-Castro (2023), addressing RQ1. While clustering analysis offers a broad overview, deeper content analysis, case studies, and user experiences are needed for clearer insights into platform diversity and applications.

5. Discussion

This study addresses three research questions (RQs) aimed at bridging key knowledge gaps in the literature on DPPs. Rather than evaluating their real-world performance, the analysis focuses on how DPPs are framed and conceptualized across academic and grey sources. Section 5.1 explores how DPPs are positioned in relation to ecosocial transitions, particularly through CI strategies. Section 5.2 examines key design aspects across the DPP lifecycle, identifying reported enablers and barriers drawn from literature. Section 5.3 analyzes the affordances of geospatial DPPs, focusing on technological features, interaction modes, and data practices. Together, these sections offer a synthesis of conceptual framings and emerging trends, clarifying how DPPs are positioned as tools for participatory action and ecosocial transition.

5.1. Framing DPPs for eco-social transformation

In addressing RQ1, this section explores how DPPs are framed as contributing to ecosocial transitions across diverse contexts. Although we do not assess actual performance, our analysis of platform descriptions reveals how they are positioned in the literature as tools for environmental awareness, inclusive urban transformation, and collective action. These conceptual framings provide valuable insights into the perceived potentials of DPPs in facilitating transition processes.

Findings confirm that DPPs evolve based on local contexts and pressing issues (Kahila-Tani et al., 2019). Falco and Kleinhans (2018) classify DPP goals as short-term practical and long-term future-oriented. Beyond pandemic responses, DPPs have addressed climate challenges through collaborative sensing, learning, and acting. Ecosocial transitions can be supported via DPPs, but the success of CI relies on motivation, engagement, and inclusivity to bridge digital divides (Arniani et al., 2015). A key takeaway is their potential to spark social innovations and promote inclusive solutions for vulnerable groups, including seniors, disabled individuals (Morelli, 2015), the homeless, refugees, ethnic minorities, and LGBT+ individuals.

For instance some examples emphasize participatory design for inclusive cities:

- The MIT Finding Places project (using interactive CityScope stations) focused on **refugees**.
- The UCL DeCID project (co-designing social infrastructure) targeted **children affected by displacement.**
- Crowd2Map fights against Female Genital Mutilation (FGM) by mapping **rural** Tanzania.
- Ushahidi was developed to map **post-election violence** in Kenya.
- Her City **empowers girls** to redesign spaces using a toolbox with the Kobo mapping app for data collection.

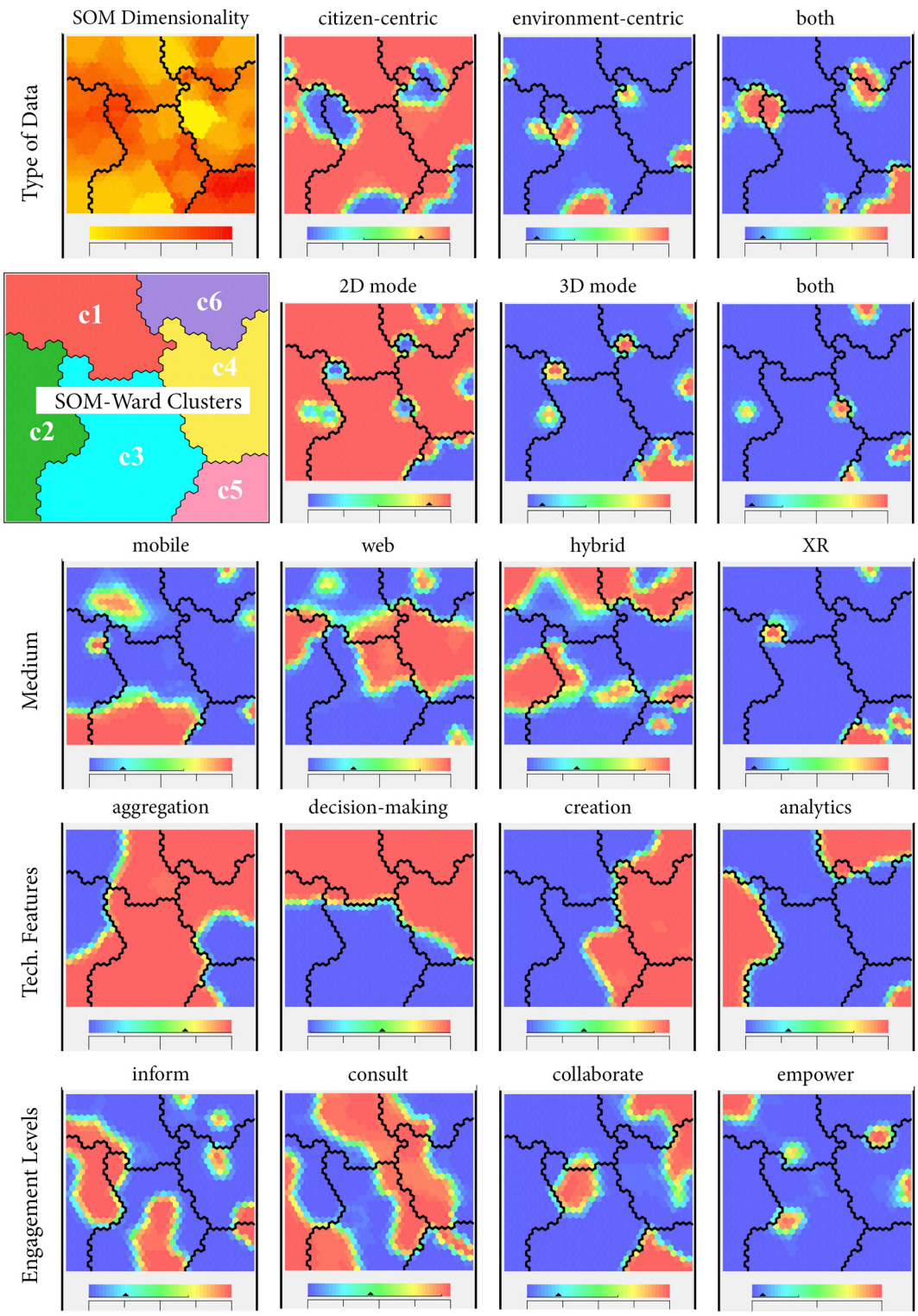

Fig. 5. SOM-Ward clustering results for several attributes.

- Limitless GIS supports **people with disabilities** by enabling them to report accessibility issues and improve infrastructure.
- The OGITO maptable application, initially developed for community mapping project in Indonesia (Aguilar et al., 2021), was adapted for two case studies. In Herne, Germany, digital map tables were used to co-design noise reduction measures, while in Zwolle, Netherlands, they facilitated planning accessible public spaces for **people with disabilities.**
- Maptionnaire supports inclusive cities through geobudgeting, enabling fair allocation of resources based on community input.

Another key takeaway emerges from citizen science approaches that leverage collective intelligence to tackle ecological concerns, with most of these platforms categorized under C3 in the SOM-Ward analysis.

- CoastSnap uses fixed cradles for crowdsourced shoreline photos to monitor **coastal changes.**

- SeaSketch facilitates collaborative spatial planning and strengthens **coastal resilience** with geospatial tools.
- SocioCoast focuses on coastal ecosystems, using web and mobile platforms to promote **sustainable tourism and biodiversity** monitoring under the Blue Flag system.
- UnionYouth engages Generation Z in environmental awareness, climate action capacity-building, and **participatory greening** to transform public spaces.
- Citclops employs Secchi disks and sensors to assess **water quality.**
- MapKnitter supports collective mapping to measure the **impact of spills** using aerial photos captured from balloons, kites, or other sources.
- FutureCoast leverages collaborative storytelling in an alternate reality game to spark discussions on **climate change.**
- CivicAI advances Augmented Collective Intelligence (ACI) to drive community transitions for **climate crisis** mitigation and adaptation, exemplified by cases like Connected Urban Forest.
- SOCIO-BEE tackles **urban air pollution** through wearable sensors and community-driven initiatives, encouraging **sustainable behaviors**.

These platforms combine data-driven monitoring, storytelling, and participatory design to address environmental challenges and advance sustainability. Evidence from socially inclusive and environmentally sustainable cities demonstrates DPPs' contributions to ecosocial transitions in both practice and theory.

The city is a municipal organization as wel as a geographic, social, and economic entity. Sennett (2013) argues that separating practical and institutional spheres diminishes their potential, while Huybrechts et al. (2017) highlight the role of participatory design in shaping shared spaces and institutions. Moore et al. (2015) propose scaling strategies—scaling out (reaching more people), scaling up (influencing laws), and scaling deep (shaping culture)—but emphasize that no fixed formula ensures success. For ecosocial transitions, DPP-enabled innovations must interconnect the physical and institutional domains of the public sphere to achieve systemic impact.

5.2. Lifecycle and key design aspects

Treisman et al. (2016) argue that DPPs have fallen short of realizing the democratic ideals of the open government movement or achieving meaningful governmental transformation. Enhancing their efficiency, effectiveness, and democratic impact (Scott et al., 2016) in PD&P requires understanding their barriers and enablers (B&E). Recent findings on B&E, analyzed through lifecycle, structural, and procedural aspects, provide a practical framework to inform key design strategies.

Kleinhans and Falco (2022) identify key challenges for DPPs, including access and awareness, sustaining motivation, expectation management, re-establishing routines, and offline follow-up and decision-making. Similarly, Royo et al. (2020) highlight barriers such as poor public knowledge, low adoption, the digital divide, lack of engagement, regulatory constraints, and failure to influence decision-making. DPPs face lifecycle challenges, from pilot phases to achieving systemic change (Murray et al., 2010). Some initiatives fail to sustain impact, while others struggle with scaling or adapting to political and cultural contexts. Whittington (2022) proposes a three-level framework to address B&E—focusing on delivering democratic innovations, expanding participation, and embedding practices into institutional processes. Each level includes B&E related to people, process, and technology (Table 2).

In the B&E framework, barriers at the Expand level include digital divides, often addressed through on-site promotion. EMPATIA used a dual survey design across four pilots (Milan, Wuppertal, Říčany, and Lisbon), analyzing 15,000+ questionnaires. Results show that hybrid approaches—combining digital and in-person engagement—improve participant diversity. Our review also confirms that DPPs play a complementary role, particularly in the initiation and evaluation phases of the planning process (Kahila-Tani et al., 2019). Paper2GIS improves accessibility in participatory mapping by digitizing paper map mark-ups, addressing mobility, visual, and literacy barriers. During COVID-19 lockdowns in the UK, it collected data via postal delivery to 525 households and was later used in Kenyan informal settlements to overcome technology access challenges (Denwood et al., 2022). The digital divide and user motivation is also key factors in SUS and TAM analysis. For example, Beta.Speckle, an early version of Speckle, tested sharing a parametric design space with citizens. Knecht et al. (2019) evaluated its usability at 66, placing it at the lower end of acceptability, and it was perceived as more suitable for experts. The tool now serves as a collaborative data hub.

Trust and ethical concerns—including privacy risks, data control, and power centralization—pose key barriers under Deliver and Embed. Harari (2021) warns of mass surveillance and digital dictatorship, advocating for (1) preventing misuse, (2) ensuring transparency, and (3) independent oversight. A Horizon project promotes privacy-enhanced sharing and participatory governance through DECODE tools (Bass and Old, 2020).

A critical barrier is the decline of platforms due to reduced usage, outdated features, or lack of sustained support (Royo et al., 2020). Decide Madrid, an award-winning e-participation platform, engaged 91,646 participants in 2018 to allocate a €100 million budget. However, participation fell to 49,535 in 2021, following a government change and a budget cut to €50 million. Recent data shows minimal activity, highlighting challenges in long-term sustainability (Olaizola and Grasso, 2023). The enabler is the commitment of future governments to sustain and improve platforms based on lessons learned from continuous feedback. Zhao et al. (2023), using Propensity Score Matching (PSM) and Difference-in-Differences (DID), showed that Sacramento 311′s 2020 updates—with real-time tracking and location data—enhanced transparency and citizen engagement.

Policymakers should prioritize open-source platforms to improve time and cost-efficiency. meinBerlin, operated by the State of Berlin, is built on the Adhocracy+ framework by Liquid Democracy. Like Decidim, it is part of a broader open-source DPP ecosystem supporting participatory governance. Currently, Adhocracy+ serves 370 organizations and 27,000 users, while Decidim supports 148 organizations and 925,152 participants worldwide.

At the Embed level in the People category, a key barrier is institutional reluctance to share real power. Ertiö (2015) reviewed 35 mobile participatory apps, finding most were not citizen-centric, focusing instead on one-way information about land-use and zoning (e.g., PlanningVIC, Master Plan 2008 Singapore, Zoner). Only a few, like Cycle Tracks, Textizen, and Commonplace, supported dialogue and strategic input. Ertiö also highlighted the lack of sustained dialogue between local governments and citizens via mobile DPPs. However, Gil et al. (2019)noted a shift from one-way information to two-way feedback becoming the new standard in recent cases.

A key barrier at the Expand level in the People category is lack of motivation to engage. Gamification is widely used to address this, as seen in Block by Block, which engaged 30,000+ participants globally—women, children, elders, refugees, and the disabled—through gamified crowdsourcing workshops. Additional strategies include place-making, follow-up mechanisms, community building, and competitive crowdsourcing. Nesta's Challenge Works and the MIT Climate CoLab used competitions to drive participation and engagement. Project Sidewalk facilitates virtual sidewalk audits to assess accessibility issues via a gamified interface with mission-based tasks, progress dashboards, and economic incentives.

The interdependent lifecycle stages—planning, development, engagement, execution, outputs, evaluation, and outcomes—enhance the success of participatory platforms, while iterative feedback continuously refines their design and operation (Treisman et al., 2016). While structural design focuses on the facilitative features and technical components, procedural design covers the processes for user engagement, and

Table 2
A framework for advancing democratic innovations (based on Whittington, 2022).

		BARRIERS	ENABLERS
Deliver	People	Traditional culture and risk aversion	Gain senior and multi-departmental buy-in
	1-2	Shortage of institutional skills	Build skills and capabilities
	Process	Rigid regulation and processes	Start small and focused, then scale
	1-2	No impact and poor communication	Transparent processes and communication
	Technology	Diffuculty commisioning and integrating digital tools	Shape new business and operating models
	1-2	Low trust in and usability of tech	Design for digital
Expand	People	Lack of diversity in participants and institutions	Use independent and diverse facilitators
	3-4	Lack of motivation to engage	Develop inclusive incentives
	Process	Processes that some citizens cannot access	Design for and with the community
	3-4	Processes that exclude some citizens	Design for increased access
	Technology	Digital exclusion	Treat inclusivity as standart
	3-4	Technology amplifying negative interactions	Design technology for community building
Embed	People	Reluctance of institutions to share real power	Grow citizen-centred cultures
	5-6	Partisanship and outsourcing	Strenghten community organising
	Process	Short-term and siloed processes	Embed participation into existing policy processes
	5-6	Compliance without conviction	Design new citizen-led institutions
	Technology	Poor development and maintanence of digital tools	Use technology as a tool for transformation
	5-6	Transparency and security risks	Build open source tools and a civic-tech community

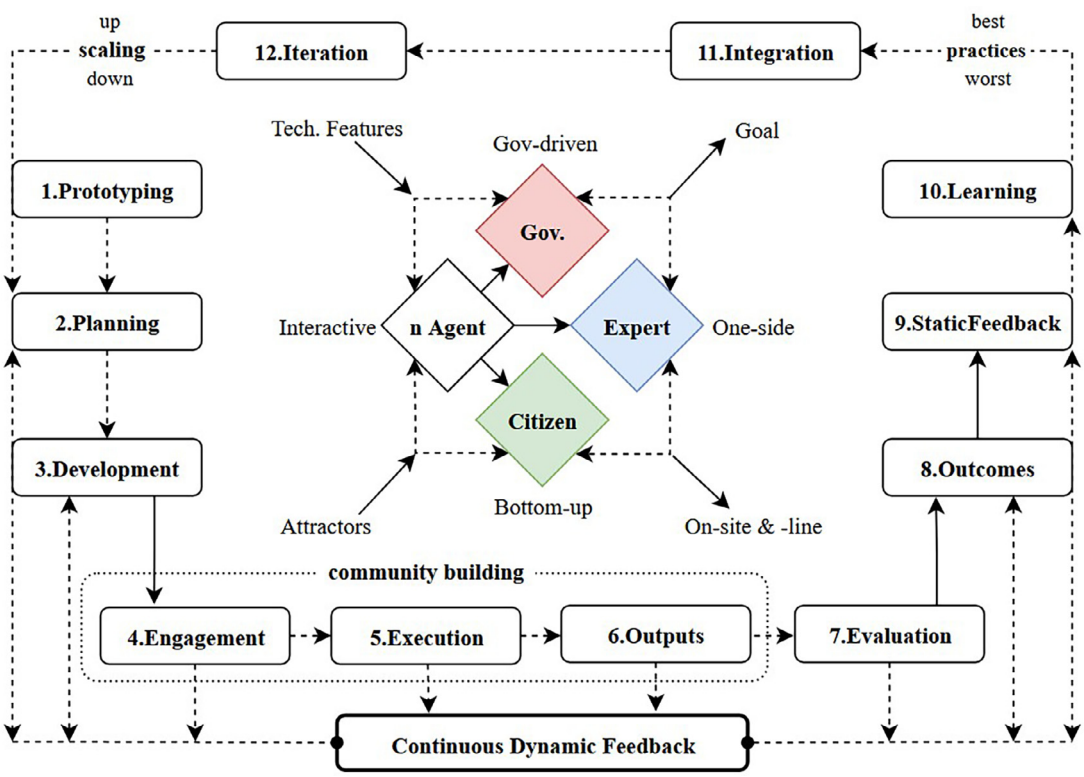

Fig. 6. Lifecycle and design framework for DPPs (adapted from Desouza and Bhagwatwar (2014)) and Treisman et al. (2016).

its possible future refinement with a public intent in Treisman' framework (Fig. 6).

Treisman et al. (2016) highlight success indicators such as participation, diversity, behavioral changes, capacity building, public impact, and future engagement. Saunders and Mulgan (2017) recommend eight principles for governing cities with CI: ensure ethics, focus on problems, reuse knowledge, select the right crowd, keep it simple, integrate processes, secure policy adoption, and address both virtual and physical worlds. The B&E and design frameworks provide a guide for deploying successful DPPs in PD&P.

5.3. Affordances of geospatial DPPs

Cross-edge technologies, including ludic, interactive, generic, sentient, immersive, and AI-powered crowdsourcing applications, emerge as distinct features in our findings, forming a cluster (C5). While highlighting technological diversity, this review also underlines the added value and potential of such tools in PD&P.

- Key features of DPP-led co-production include collaborative mapping, comments, idea submission, voting, ranking, media uploads, and analytics (Falco and Kleinhans, 2018). Tools are evolving into AI-powered actions and interactive features like XR scenarios and algorithm training (CivicAI) for responsive engagement. For instance, Sidewalk AI (Morra et al., 2024) uses computer vision and predictive analytics to process crowdsourced data, detecting width, obstacles, and pavement conditions to enhance urban accessibility. 4D Citymaking integrates projection mapping, AR, object tracking, and touchscreens to support collaborative decision-making and visualize urban futures. DIPAS combines CityScope and online tools with AI

analytics and mobile touch tables for feedback processing and scenario simulation (Lieven et al., 2020).

- Network-based approaches like Architasker offer greater interactivity than contest-based crowdsourcing, like Arcbazaar, supporting participatory design through iterative micro-tasks by architects and non-experts. However, both may face cost-efficiency concerns due to unselected proposals.
- Crowdsourcing DPPs are widely used to explore multi-scale city dynamics. UrbanOpticon maps collective mental images, StreetScore assesses streetscape safety, and Place Pulse evaluates perceptions of safety and livability. Arturo analyzes habitability across 10% of Madrid's streets using 50 parameters. Platforms like Streetchange, Desirable Streets, Future Mobility, Tourism4Cities, Citibeats, and SocialGlass use web scraping to analyze social media, pedestrian movements, and vehicle flows for responsive decision-making. While these tools leverage AI analytics to visualize citizen-generated data, participation often remains limited to analytics or even data extraction, falling short of citizen-centered empowerment. Within the Collective Awareness Platforms for Sustainability and Social Innovation (CAPS) ecosystem, it can be categorized into two distinct approaches, delineated by the participants' levels of empowerment and (un)awareness: aggregation and co-creation (Arniani et al., 2015). Peach et al. (2021) stress diversity, free idea-sharing, data integration, and citizen-centered empowerment over data extraction.
- XR-based DPPs enhance immersive engagement in urban design. Habla fosters citizen to citizen dialogue about public spaces, while Veus supports voting, feedback, and suggest improvements to governments. CoDAS enables real-time visualization for collective decision-making, while AvaLinn links long-term visions with current actions. A recent review on participatory XR highlights technical issues, inclusiveness, and adoption as key challenges, while emphasizing visual realism and real-time modifications as major benefits (Simonofski et al., 2024).
- Gamified strategies are increasingly prominent in DPPs. Superbarrio enables users to assess ecological, economic, and social impacts of decisions in 3D spaces, while EquiCity, a multiplayer generative game, balances sustainability goals through mathematical frameworks and simulations. Qurbancraft revitalizes underused spaces by allowing users to design and vote on modular layouts. Bai et al. (2018) advance the Urban Gallery methodology using AR for real-time feedback and a grid-based approach to balance controversies enabling data-driven, inclusive, and adaptable urban design through interactive gaming.

Pioneering research-based DPPs offer diverse human-computer interactions and advanced techniques, including interactive tables (MIT's CityScopes, DIPAS, 4D Citymaking), AR/MR applications (Habla, Veus, Urban CoBuilder, CommunitAR), parametric design tools (Beta.Speckle, EquiCity), urban games (MUV, GeoGame B3, SuperBarrio, SuperLupa), and crowd-creative spaces (Qua-kit, QurbanCraft). Environmental analytics leverage bio-optical sensors (Citclops/EyeonWater), CoastSnap's in situ cradles, wearables, drones (SOCIO-BEE), and low-cost IoT sensors (DECODE pilots). Additionally, intelligent PSSs empower users to reshape environments and contextualize data through ubiquitous computing technologies (Deal et al., 2017, Pak, 2017).

6. Conclusion

This study provides a comprehensive analysis of digital participatory platforms (DPPs), offering a global inventory of 312 mobile and web-based platforms and addressing critical knowledge gaps in the literature. Rather than evaluating real-world performance or efficiency of these platforms, the analysis focuses on identifying recurring conceptual framings, thematic patterns, and key design attributes. To this end, the study employs a descriptive, interpretive, and exploratory approach, grounded in a systematic review of documented cases.

By combining a three-step systematic review and AI-powered clustering technique, we identified patterns, features, and affordances across these platforms, emphasizing their role in ecosocial transitions. While previous studies primarily relied on SWOT, thematic, or typology-based analyses, our SOM-Ward clusters introduce a novel approach for meta-analysis of the ever-evolving DPP landscape. By integrating Participatory Action Research (PAR) and Collective Intelligence (CI) approaches, this study also develops a CI genome to deepen insights into platform affordances. While it is not presented as a fixed model, the proposed genome provides an adaptable analytical framework, offering conceptual and methodological flexibility for future research.

Our findings highlight a shift from interpretative flexibility to 'closure' (Renteria and Varela-Castro, 2023) in DPP innovations, particularly in government-driven geo-budgetting, data-centered consultative platforms. Although cross-edge technologies—such as interactive, immersive, and AI-powered applications—show promise in advancing participatory urban planning, DPPs still face technical and non-technical challenges throughout their lifecycle, from designing to embedding.

To address these challenges, we propose a lifecycle framework and outline key design aspects—including accessibility, transparency, and scalability—to guide the development of successful DPPs. This research contributes to sustainable urban governance by offering practical strategies for policymakers, designers, and civic initiatives to effectively harness these platforms for ecosocial transitions. It underscores the need to strengthen both online and onsite citizen participation and leverage enablers to enhance real-world applications and achieve systemic impact.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Muhammet Ali Heyik: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Meral Erdoğan:** Writing – review & editing, Supervision. **José María Romero-Martínez:** Writing – review & editing, Supervision, Project administration.

Acknowledgement

This research was conducted as part of the EU project EMCROTUR: Emergencias Crónicas y Transformación Ecosocial en Espacios Costeros Turistificados. It was co-funded by the Spanish Ministry of Science and Innovation and the European Regional Development Fund (ERDF) under project reference PID2022-137648OB-C22. The authors would like to acknowledge that this paper is submitted in partial fulfilment of the requirements for PhD degree at Yildiz Technical University.

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